Autonomous Satellite Tracking System Using TLE Data

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1 Abstract

This project presents a complete software-based autonomous satellite tracking system. The system utilizes orbital data derived from TLE (Two-Line Element) sets to predict satellite motion, evaluate visibility conditions, and determine the optimal satellite pass for tracking. The system also computes the required pointing angles for telescope control.

2 Introduction

Satellite tracking is an essential component in space applications, including communication, Earth observation, and research. This project focuses on developing an intelligent tracking system that autonomously selects the best satellite to track based on multiple criteria such as visibility duration, elevation angle, and distance from the observer.

3 Methodology

3.1 Orbit Prediction

The satellite orbits are calculated using TLE data and propagated to obtain position coordinates in 3D space.

3.2 Pass Evaluation

Each satellite pass is analyzed using:

- Visibility duration
- Maximum elevation angle
- Minimum distance from the observer

3.3 Tracking Score

A scoring system is implemented to select the best satellite:

$$Tracking\ Score = (Max\ Elevation \times 2) - (Distance/1000) + (Visible\ Points \times 0.5)$$

4 Results and Analysis

4.1 Selected Orbit

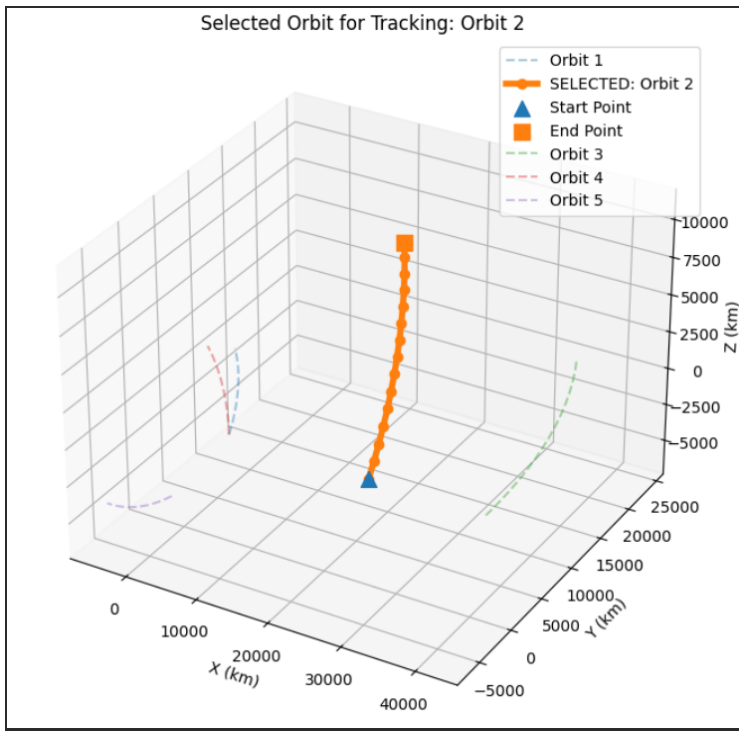

Figure 1: Selected orbit for tracking based on the highest tracking score.

4.2 Motion Direction

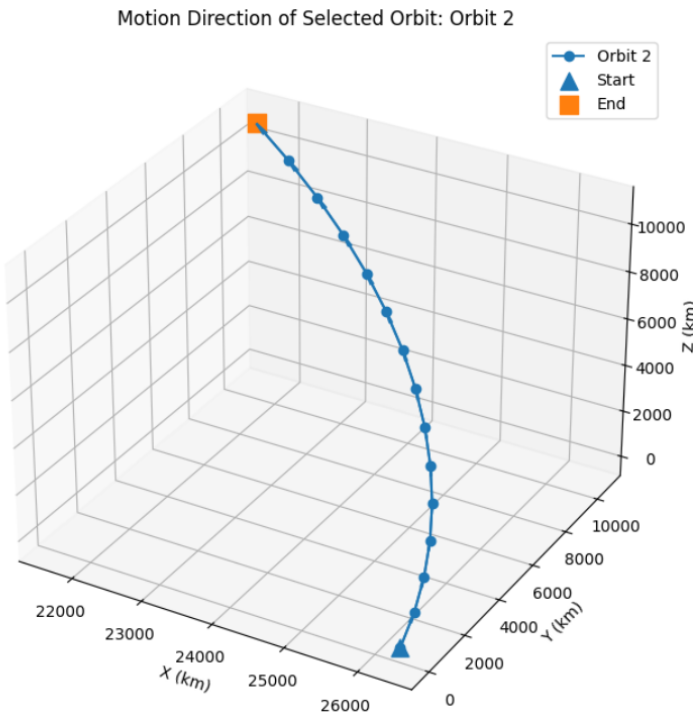

Figure 2: Motion direction of the selected satellite orbit, showing start and end points.

4.3 Tracking Decision Analysis

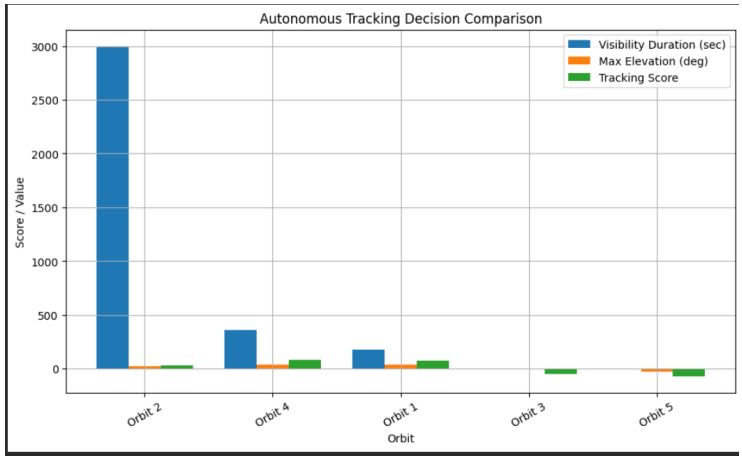

Figure 3: Comparison between multiple satellite orbits using tracking score metrics.

4.4 Telescope Pointing Angles

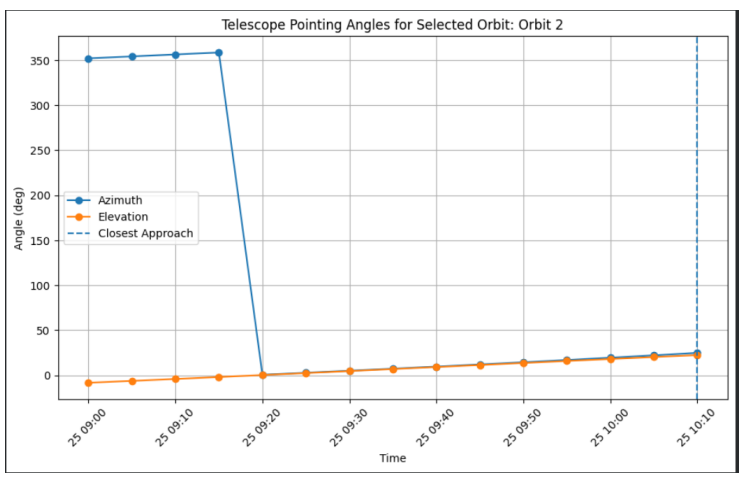

Figure 4: Azimuth and elevation angles required for telescope tracking.

4.5 3D Orbit Visualization

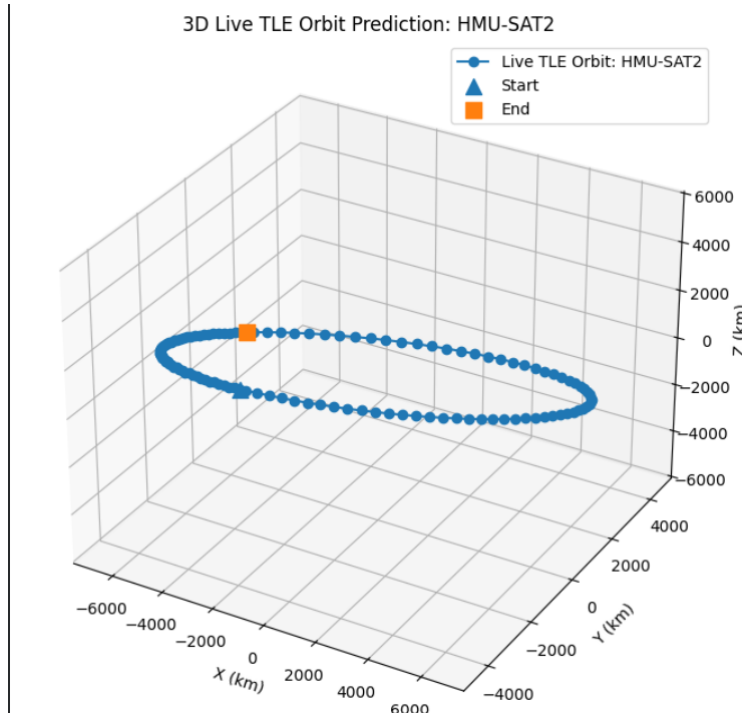

Figure 5: 3D visualization of satellite orbit using real-time TLE data.

5 Discussion

The results demonstrate that the system can successfully evaluate multiple satellite passes and select the optimal one for tracking. The motion direction and pointing angles provide all necessary data for real-world telescope control. The scoring system ensures that the selected satellite has the best visibility and tracking conditions.

6 Conclusion

This project successfully developed an autonomous satellite tracking system using only software tools. The system integrates orbit prediction, pass evaluation, decision-making, and angle computation. It provides a complete solution for satellite tracking applications and can be extended to real-time hardware implementation.